

HANNAH KAMEN

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PROFILE

Applied environmental and resource economist. My research investigates how environmental risk and local land use regulation reshape urban housing markets and the welfare of the households who live in them. Published in peer-reviewed journals and a forthcoming USDA Economic Information Bulletin.

EDUCATION

Ph.D., Mineral & Energy Economics (Applied Economics) <i>Colorado School of Mines, Golden, CO</i>	Expected Fall 2026
M.S., Mineral & Energy Economics (Applied Economics) <i>Colorado School of Mines, Golden, CO (GPA: 3.92)</i>	2022
B.A., Economics (with Distinction) <i>University of Rochester, Rochester, NY</i>	2015

RESEARCH FIELDS

Environmental and Resource Economics; Urban Economics; Housing and Land Use Policy; Applied Econometrics and Causal Inference.

WORKING PAPERS

“Can Residential Zoning Reform Mitigate Climate Gentrification? Spatial Sorting on Urban Microclimates.” *Job Market Paper.*

I build a structural equilibrium model of household location choice to evaluate how public cooling infrastructure interacts with local zoning in heat-vulnerable, low-income neighborhoods. I find that Phoenix’s Cool Corridors program is regressive on its own, benefiting low-income households negligibly, but that pairing it with spatially targeted upzoning reverses this gradient, raising their gains to \$506 annually. I further show that the homothetic demand specification standard in the environmental sorting literature systematically understates the welfare loss that cooling-induced rent increases impose on low-income households.

“The Price of Cool: How Urban Heat Shapes Neighborhood Value.”

I develop a novel instrumental variable approach based on fine-scale urban airflow to identify household willingness to pay for cooling at multiple distances from the home, using residential sales data from Maricopa County, AZ paired with 1km-resolution NCAR urban air temperature data. I show that households pay \$676 annually for a 1°C reduction in summer temperatures within a few blocks of the home and \$245 for equivalent cooling at the broader neighborhood scale. I quantify the benefits of Phoenix’s Cool Corridors program at \$137M and show that conventional tract-level evaluations understate program value by 53%.

“Extreme Heat and U.S. Labor Relocation: Implications for Inequality.”

I build a linked discrete choice and large-scale computable general equilibrium (CGE) optimization framework to forecast 30-year wage and housing market impacts of climate-driven heat exposure across U.S. regions, modeling how household location, employment, and wages adjust to changing climate risk.

PUBLICATIONS

- Winikoff J.B., **H. Kamen**, R.W. Winkler (2026). “Household Energy Use in Rural America: Costs, Burdens, and Contributing Factors.” Forthcoming USDA Economic Information Bulletin.
- Carbone J.C., L. Jiang, B. O’Neill, Y. Shen, H. Zoraghein, and **H. Kamen** (2026). “How Will Climate-Induced Migration Affect the Future Distribution of Population in the US?” Revisions requested at Nature Climate Change.
- Wobus C., E. Small, J.C. Carbone, P. Modi, **H. Kamen**, W. Szafranski, B. Livneh (2022). “Water Allocation, Return Flows, and Economic Value in Water-Scarce Environments.” *Water* 14(20):3280.
- Kuwayama Y., **H. Kamen** (2016). “What Drives the Reuse of Municipal Wastewater? A County-Level Analysis of Florida.” *Land Economics* 92: 679–702.

INVITED TALKS/PRESENTATIONS

Occasional Workshop in Environmental and Resource Economics (2026); Camp Resources XXXII (2026); North American Meetings of the Regional Science Association International (2024); Berkeley-Sloan Summer School

(2023); American Geophysical Union Fall Meeting (2021, 2022); AGU Chapman Conference (2022); University of Colorado Environmental & Resource Economics Workshop (2022)

AWARDS AND FELLOWSHIPS

William Jesse Coulter Award in Mineral and Energy Economics, Spring 2026; Carol Dahl Fellowship, Fall 2024 – Spring 2025; Pauli Scholarship, Spring 2024; Berkeley-Sloan Summer School Fellowship, 2023; University of Rochester Arts, Sciences & Engineering Merit Scholarship, 2011–2015; University of Rochester Dean’s List 6/8 eligible semesters, 2011–2015; American Groundwater Trust Amtrol Scholarship, 2011; Intel Semifinalist Award, 2011.

RESEARCH EXPERIENCE

Grid Planning Researcher, National Renewable Energy Laboratory (NREL) 2024–2026
Golden, CO

Contributed to NREL’s flagship U.S. electricity capacity-expansion optimization model, developing and testing scenarios in Python and GAMS to quantify the impact of Inflation Reduction Act tax credits on renewable energy deployment.

Economist Student Trainee (Intern), USDA Economic Research Service Summer 2024
Resource & Rural Economics Division, Kansas City, MO

Produced statistical analysis and econometric modeling for a forthcoming USDA Economic Information Bulletin on household energy cost burden and rural housing disparities. Cleaned and analyzed large public micro-datasets in Python and Stata, authored sections of the publication, and produced GIS maps and visualizations translating findings for policymakers and general audiences.

Research Assistant, NSF Stream Systems Program 2020–2024
Colorado School of Mines, Golden, CO

Built a structural GAMS-based optimization model of regional water trade with an interdisciplinary team of economists, hydrologists, and climate scientists, quantifying the economic value of water allocation and return flows across drought-prone western U.S. basins. Presented findings independently at academic conferences.

Research Intern, Resources for the Future (RFF) Summer 2014
Environmental Economics Think Tank, Washington, DC

Supported a team of environmental economists on a study of municipal water-reuse adoption across Florida counties; contributed to data collection, econometric modeling, and drafting that led to a co-authored peer-reviewed publication in Land Economics.

PROFESSIONAL EXPERIENCE

Data Scientist (Team Lead, Senior Analyst, Analyst, Ad Manager) 2016–2020
SocialCode (now Code3), New York, NY

Led the data science research function at a digital advertising agency, designing and running the team’s causal measurement program (incrementality testing, geo-based lift, media mix modeling) and building regression and machine learning models on large auction-platform datasets across YouTube, Google Search, and Facebook/Instagram.

Research Associate, National Economic Research Associates (NERA) 2015–2016
Antitrust & Competition, New York, NY

Produced quantitative economic and statistical analyses supporting expert testimony in DOJ and FTC antitrust matters using SQL and SAS on large transactional datasets.

TECHNICAL SKILLS

Econometrics & modeling: Causal inference (instrumental variables, difference-in-differences), quasi-experimental design, discrete choice and structural equilibrium modeling, panel data and time-series methods, large-scale optimization.

Spatial analysis & GIS: Parcel-level spatial data construction, GIS mapping (geopandas, QGIS), spatial econometrics, integration of high-resolution environmental data (NCAR HUMID, NOAA HYSPLIT) with administrative records.

Programming & data: Python (pandas, numpy, statsmodels, scikit-learn, geopandas, matplotlib, Bokeh), R, SQL, Stata, SAS, GAMS, Git/GitHub.